Debiased Estimators in High-Dimensional Regression

A Review and Replication of Javanmard and Montanari (2014)

Benjamin Smith [1]

[1]Department of Statistical Sciences
University of Toronto

STA3000 Final Presentation

Statistical Sciences
UNIVERSITY OF TORONTO

Background

- Modern statistical problems are widely recognized as having become increasingly high-dimensional ($p \gg n$)

  Examples

  - Genomics
  - Neuroimaging
  - Asset Pricing
  - Signal Processing
  - Collaborative Filtering

- LASSO regression (Tibshirani, 1996) has become a widely applicable and successful technique for addressing such problems.

Background

- We work in a linear regression model setting where n i.i.d. pairs $(Y_1, X_1), (Y_2, X_2), \ldots, (Y_n, X_n)$ are given with vectors $X_i \in \mathbb{R}^p$ and response variables defined as

$$Y_i = \langle \theta_0, X_i \rangle + W_i, \quad W_i \sim N(0, \sigma^2) \tag{1}$$

  Where $\theta_0 \in \mathbb{R}^p$ and $\langle \cdot, \cdot \rangle$ is the standard scalar product in $\mathbb{R}^p$.
- In matrix form

$$Y = \mathbf{X}\theta_0 + W, \quad W \sim N(0, \sigma^2 I_{n \times n}) \tag{2}$$

Background

- The LASSO produces sparse estimates by incorporating an ℓ_1 penalty into the objective function:

$$\hat{\theta}^n(Y, \mathbf{X}; \lambda) = \arg\min_{\theta \in \mathbb{R}^p} \left\{ \frac{1}{2n} \| Y - \mathbf{X}\theta \|_2^2 + \lambda \|\theta\|_1 \right\} \tag{3}$$

- **The Catch:** The ℓ_1 penalty used in LASSO shrinks coefficients toward zero, introduces bias into the estimators!

The Problem With LASSO

- LASSO encourages sparse models - but introduces bias that complicates statistical inference!
- The resulting sampling distribution of the selected model is difficult to characterize.
- i.e. Its not possible to produce appropriate
  - Confidence Intervals
  - Hypotheses Tests
  - P-values

With the current construction (3)

The Solution: Javanmard and Montanari (2014)

Journal of Machine Learning Research 15 (2014) 2869-2909 Submitted 3/14; Revised 4/14; Published 10/14

Confidence Intervals and Hypothesis Testing for High-Dimensional Regression

Adel Javanmard ADELJ@STANFORD.EDU
Department of Electrical Engineering
Stanford University
Stanford, CA 94305, USA

Andrea Montanari MONTANAR@STANFORD.EDU
Department of Electrical Engineering and Department of Statistics
Stanford University
Stanford, CA 94305, USA

- Javanmard and Montanari (2014) propose a debiasing procedure for LASSO that allows for traditional statistical inference!
- How?

Javanmard and Montanari (2014): Debaising Algorithm

Algorithm 1 Unbiased estimator for θ_0 in high-dimensional linear regression models

Input: Measurement vector y, design matrix $\mathbf{X}$, parameters λ, μ.
Output: Unbiased estimator $\hat{\theta}^u$.

1: Let $\hat{\theta}^n = \hat{\theta}^n(y; \mathbf{X}; \lambda)$ be the LASSO estimator as per (3).
2: Set $\widehat{\Sigma} = (\mathbf{X}^\top \mathbf{X})/n$.
3: **for** $i = 1, 2, \ldots, p$ **do**
4: Let m_i be a solution of the convex program:

$$\begin{aligned} &\text{minimize} && m^\top \widehat{\Sigma} m \\ &\text{subject to} && \|\widehat{\Sigma} m - e_i\|_\infty \leq \mu, \end{aligned} \tag{4}$$

where $e_i \in \mathbb{R}^p$ is the vector with one at the i-th position and zero everywhere else.
5: **end for**
6: Set $M = (m_1, \ldots, m_p)^\top$. If any of the above problems is not feasible, then set $M = I_{p \times p}$.
7: Define the estimator $\hat{\theta}^u$ as follows:

$$\hat{\theta}^u = \hat{\theta}^n(\lambda) + \frac{1}{n} M \mathbf{X}^\top \big(Y - \mathbf{X}\hat{\theta}^n(\lambda)\big) \tag{5}$$

- **Basic Intuition:** the subgradient of the ℓ_1 norm at the LASSO solution $\hat{\theta}^n$ is $\mathbf{X}^\top(Y - \mathbf{X}\hat{\theta}^n)/(n\lambda)$.
- By adding a term which is proportional to the subgradient, (5) compensates the bias introduced by the ℓ_1 penalty in the LASSO.

Presentation Schedule

1. **Prerequisites:** Compatibility Constant & Generalized Coherence
2. **The Generalized Debiased Estimator ($\hat{\theta}^*$):** Error Decomposition & Guarantees
3. **Asymptotics & Error Bounds** for $\hat{\theta}^u$
4. **Valid Inference:** Confidence Intervals, Testing, & Robustness
5. **Empirical Analysis & Extensions:** Simulation Replications & Riboflavin Data + Lasso Projection (Zhang and Zhang, 2014)
6. **Discussion & Conclusions**

Prerequisites

Definition: Compatibility Constant

Given a symmetric matrix $\hat{\Sigma} \in \mathbb{R}^{p \times p}$ and a set $S \subseteq [p]$ ($[p] = \{1, \ldots, p\}$ the index set of all parameters), the corresponding **compatibility constant** is defined as

$$\phi^2(\hat{\Sigma}, S) \equiv \min_{\theta \in \mathbb{R}^p} \left\{ \frac{|S| \langle \theta, \hat{\Sigma}\theta \rangle}{\|\theta_S\|_1^2} : \theta \in \mathbb{R}^p, \|\theta_{S^c}\|_1 \leq 3\|\theta_S\|_1 \right\} \tag{6}$$

- $\hat{\Sigma} \in \mathbb{R}^{p \times p}$ satisfies the compatibility condition for $S \subseteq [p]$ with constant ϕ_0 if $\phi(\hat{\Sigma}, S) \geq \phi_0$.
- We say it holds for the design matrix $\mathbf{X}$ if it holds for $\hat{\Sigma} = \mathbf{X}^\top \mathbf{X}/n$.
- **The Compatibility Constant measures if the true signal θ_S stands out distinctly from the "noise" in the design matrix $\mathbf{X}$.**

Prerequisites

Definition: Generalized Coherence Parameter

Given the pair $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $M \in \mathbb{R}^{p \times p}$. Let $\hat{\Sigma} = \mathbf{X}^\top \mathbf{X}/n$ denote the associated sample covariance. The generalized coherence parameter of $\mathbf{X}, M$ is:

$$\mu_*(\mathbf{X}; M) \equiv |M\hat{\Sigma} - I|_\infty = \max\left\{\max_i |(M\hat{\Sigma})_{ii} - 1|, \max_{i \neq j} |(M\hat{\Sigma})_{ij}|\right\} \tag{7}$$

- **The Generalized Coherence Parameter measures the worst-case residual correlation after applying M, which bounds the residual bias.**
- If columns of $\mathbf{X}$ are ℓ_2-normalized to $\sqrt{n}$:

$$\mu_*(\mathbf{X}; I) = \frac{1}{n} \max_{i \neq j} |\langle \mathbf{X}e_i, \mathbf{X}e_j \rangle| \quad (e_i \text{ is the } i\text{-th basis vector}) \tag{8}$$

The Generalized Debiased Estimator ($\hat{\theta}^*$)

In order to clarify the distributional properties of $\hat{\theta}^u$ (in (5)) Javanmard and Montanari (2014) considers a general debiasing procedure that makes use of an arbitrary debiasing matrix $M \in \mathbb{R}^{p \times p}$ and define the **generalized debiased estimator** $\hat{\theta}^*$ as:

Generalized Debiased Estimator

$$\hat{\theta}^*(Y, \mathbf{X}; M, \lambda) = \hat{\theta}^n(\lambda) + \frac{1}{n} M \mathbf{X}^\top (Y - \mathbf{X}\hat{\theta}^n(\lambda)) \tag{9}$$

The Generalized Debiased Estimator ($\hat{\theta}^*$)

Theorem 6: Error Decomposition of $\hat{\theta}^*$

For design $\mathbf{X} \in \mathbb{R}^{n \times p}$ and debiased estimator $\hat{\theta}^*$:

1. **Decomposition:** $\sqrt{n}(\hat{\theta}^* - \theta_0) = Z + \Delta$
   - $Z \sim N(0, \sigma^2 M \hat{\Sigma} M^\top)$ where $Z = M\mathbf{X}^\top W / \sqrt{n}$
   - $\Delta = \sqrt{n}(M\hat{\Sigma} - I)(\theta_0 - \hat{\theta}^n)$
2. **Bound:** Under compatibility (ϕ_0) and coherence (μ_*), with $\lambda = \sigma\sqrt{\frac{c^2 \log p}{n}}$ and $K = \max \hat{\Sigma}_{ii}$:

$$\mathbb{P}\left(\|\Delta\|_\infty \geq \frac{4c\mu_* \sigma s_0}{\phi_0^2} \sqrt{\log p} \right) \leq 2p^{-c_0}, \quad c_0 = \frac{c^2}{32K} - 1$$

**If M minimizes $|M\hat{\Sigma} - I|_\infty$, then $\mu_* = \mu_{\min}(\mathbf{X})$.*

The Generalized Debiased Estimator ($\hat{\theta}^*$)

Theorem 7: Probabilistic Guarantees (Technicality)

Assume $\mathbf{X}$ has independent sub-Gaussian rows and well-behaved covariance Σ. For some constants $c_1, c_2 > 0$:

1. **Compatibility Event ($\mathcal{E}_n$):** If $n \gtrsim s_0 \log(p/s_0)$, the compatibility condition holds ($\phi \geq \phi_0$) with probability:

$$\mathbb{P}(\mathcal{E}_n(\phi_0, s_0, K)) \geq 1 - 4e^{-c_1 n}$$

2. **Feasibility Event ($\mathcal{G}_n$):** If $n \gtrsim \log p$, the optimization for M is feasible ($\mu_{\min}(\mathbf{X}) < a\sqrt{\frac{\log p}{n}}$) with probability:

$$\mathbb{P}(\mathcal{G}_n(a)) \geq 1 - 2p^{-c_2}$$

**In short: Both ϕ_0 and μ_* can be bounded with probabilities rapidly converging to 1 as $n, p \to \infty$.*

Asymptotic Normality of $\hat{\theta}^u$

Theorem 8: Error Bounds for Debiased Estimator ($\hat{\theta}^u$)

For the debiased LASSO estimator $\hat{\theta}^u$ with $\mu = a\sqrt{(\log p)/n}$:

1. **Decomposition:** $\sqrt{n}(\hat{\theta}^u - \theta_0) = Z + \Delta$
   - $Z|\mathbf{X} \sim N(0, \sigma^2 M\hat{\Sigma}M^\top)$ where $Z = M\mathbf{X}^\top W/\sqrt{n}$
   - $\Delta = \sqrt{n}(M\hat{\Sigma} - I)(\theta_0 - \hat{\theta}^n)$
2. **Tail Bound:** Under Theorem 7 assumptions, for sufficiently large n:

$$\mathbb{P}\left(\|\Delta\|_\infty \geq \left(\frac{16ac\sigma}{C_{\min}}\right)\frac{s_0 \log p}{\sqrt{n}}\right) \leq 4e^{-c_1 n} + 4p^{-\min(\tilde{c}_0, c_2)}$$

***Asymptotic Consequence:** This establishes $\|\Delta\|_\infty = \mathcal{O}(s_0 \log p/\sqrt{n})$. If sparsity satisfies $s_0 = o(\sqrt{n}/\log p)$, the bias vanishes ($\|\Delta\|_\infty = o(1)$), making $\hat{\theta}^u$ asymptotically unbiased.

Valid Inference

- Theorem 8 shows that asymptotic normality is achieved for $\hat{\theta}^u$ with vanishing bias under sparse conditions.
- Assuming the sparsity assumption to be $s_0 = o(\sqrt{n}/\log p)$ we note the following Lemma

Lemma 10: Uniform Convergence to Normal CDF

Let $\mathbf{X}$ satisfy sub-Gaussian design assumptions and let $\hat{\sigma}$ be a consistent noise estimator.
If $s_0 = o(\sqrt{n}/\log p)$, the standardized debiased estimator is uniformly asymptotically normal over all sparse vectors θ_0:

$$\lim_{n\to\infty} \sup_{\|\theta_0\|_0 \le s_0} \sup_{x\in\mathbb{R}} \left| \mathbb{P}\left(\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \le x \right) - \Phi(x) \right| = 0$$

*(*Where $\Phi(x)$ denotes the standard normal CDF. This validates coordinate-wise hypothesis testing for the debiased estimator.)*

Valid Inference: Core Results

(A) Construction of Confidence Intervals

$(1-\alpha) \times 100\%$ Confidence Interval

$$\begin{aligned} J_i(\alpha) &\equiv \left[\hat{\theta}_i^u - \delta(\alpha, n), \hat{\theta}_i^u + \delta(\alpha, n)\right], \\ \delta(\alpha, n) &\equiv \Phi^{-1}(1-\alpha/2)\frac{\hat{\sigma}}{\sqrt{n}}[M\hat{\Sigma}M^\top]_{i,i}^{1/2} \end{aligned} \tag{10}$$

It can also be shown that the confidence interval is asymptotically valid, namely

$$\lim_{n\to\infty} \mathbb{P}\left(\theta_{0,i} \in J_i(\alpha)\right) = 1-\alpha \tag{11}$$

Valid Inference: Core Results

(B) Valid Hypothesis Testing

For $H_{0,i} : \theta_{0,i} = 0$, we define the p-value and reject if $P_i \leq \alpha$:

$$P_i = 2\left(1 - \Phi\left(\frac{\sqrt{n}|\hat{\theta}_i^u|}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}}\right)\right)$$

From a formal minimax perspective, this test controls the worst-case errors over all sparse models ($\|\theta_0\|_0 \leq s_0$):

$$\textbf{Size } (\alpha_{i,n}) \equiv \sup_{\theta_0} \{\mathbb{P}_{\theta_0}(\text{Reject } H_{0,i}) : \theta_{0,i} = 0\}$$

$$\textbf{Type II } (\beta_{i,n}(\gamma)) \equiv \sup_{\theta_0} \{\mathbb{P}_{\theta_0}(\text{Accept } H_{0,i}) : \theta_{0,i} \geq \gamma\}$$

Valid Inference: Core Results

(C) Lemma: Joint Asymptotic Normality (Simultaneous CIs)

For a fixed subset R of size k, let $Q^{(n)} \equiv \frac{\hat{\sigma}^2}{n}[M\hat{\Sigma}M^\top]$. If $s_0 = o(\sqrt{n}/\log p)$, the joint distribution converges uniformly to a multivariate normal CDF ($\mathbf{\Phi}_k$):

$$\lim_{n\to\infty} \sup_{\|\theta_0\|_0 \le s_0} \left| \mathbb{P}\left\{ \left(Q^{(n)}_{R,R}\right)^{-1/2} (\hat{\theta}^u_R - \theta_{0,R}) \le x \right\} - \mathbf{\Phi}_k(x) \right| = 0$$

---

1. Valid Confidence Regions:
$J_R(\alpha) \equiv \hat{\theta}^u_R + (Q^{(n)}_{R,R})^{1/2}\mathcal{C}_{k,\alpha}$, yielding $1-\alpha$ coverage.

2. FWER Control:
Using the Bonferroni decision rule $\widehat{T}^F_{i,\mathbf{X}} = \mathbf{1}\{P_i \le \alpha/p\}$, the asymptotic FWER is tightly controlled: $\limsup_{n\to\infty} \text{FWER} \le \alpha$.

(4) Extension: Non-Gaussian Noise

For valid inference under non-Gaussian noise, the scaled error summands ξ_j must satisfy the Lindeberg condition:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} \mathbb{E}\left(\xi_j^2 \mathbb{I}_{\{|\xi_j|>\varepsilon\sqrt{n}\}} \mid X\right) = 0$$

To enforce this, we modify the debiasing matrix construction by adding an observation-wise influence constraint:

$$\begin{aligned} \underset{m\in\mathbb{R}^p}{\text{minimize}} \quad & m^\top \hat{\Sigma} m \\ \text{subject to} \quad & \|\hat{\Sigma} m - e_i\|_\infty \le \gamma \\ & \|Xm\|_\infty \le n^\beta \quad (\text{for } \beta \in (1/4, 1/2)) \end{aligned}$$

*(*This guarantees the Central Limit Theorem holds, keeping all previous tests and intervals asymptotically valid.)*

Empirical Analysis: Background

- **Replication Scope:** Evaluated 6 simulation scenarios and the riboflavin (vitamin B_2) real-data application from Javanmard and Montanari (2014).
- **Extended Benchmarking:** Compared the original method against the Lasso projection estimator (Zhang and Zhang, 2014), providing a more comprehensive baseline than the original study.
- **Computational Setup:** Executed in `R` v4.2.2 using parallel processing (`doParallel`, `foreach`) on an Intel Core i7 processor.
- **Code & Data Sources:**
  - Debiased LASSO: Authors' official repository.
  - Baselines & Dataset (Multisample splitting, Ridge/Lasso projection): `hdi` `R` package.

Empirical Analysis: Simulation Study

We consider the linear model:

$$Y = \mathbf{X}\theta_0 + W, \quad W \sim N(0, \sigma^2 I_{n\times n})$$

where the covariance $\Sigma \in \mathbb{R}^{p\times p}$ is a circulant symmetric matrix. For $j \leq k$:

$$\Sigma_{jk} = \begin{cases} 1 & \text{if } k = j \\ 0.1 & \text{if } k \in \{j+1, \ldots, j+5\} \\ & \text{or } k \in \{j+p-5, \ldots, j+p-1\} \\ 0 & \text{otherwise} \end{cases}$$

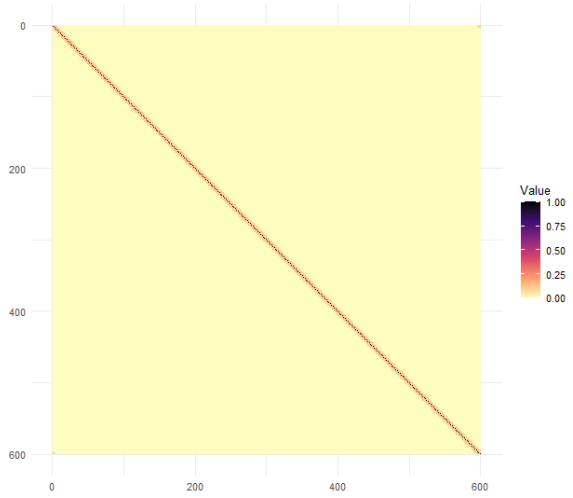


Figure: Example circulant symmetric matrix Σ with $p = 600$

- **DGP:** Random support ($|S| = s_0$), $\theta_{0,i} = b$ for $i \in S$, else 0; $W_i \sim N(0, 1)$.
- **Setup:** Multiple (n, p, s_0, b); avg. over 20 runs; default tuning (no μ specified).

Empirical Analysis: Simulation Study

- **Avg. Interval Length:** $\ell = p^{-1} \sum_{i \in [p]} \text{Avglength}(J_i(\alpha))$
- **By Support:** ℓ_S (active), ℓ_{S^c} (inactive)
- **Coverage:** $\widehat{\text{Cov}} = p^{-1} \sum_i \widehat{\mathbb{P}}[\theta_{0,i} \in J_i(\alpha)]$
- **By Support:** $\widehat{\text{Cov}}_S$, $\widehat{\text{Cov}}_{S^c}$
- **Other Metrics:** False Positives (FP), True Positives (Power)

Empirical Analysis: Simulation Study

Configuration \ Measure	ℓ	ℓ_S	ℓ_{S^c}	$\widehat{Cov}$	$\widehat{Cov}_S$	$\widehat{Cov}_{S^c}$
(1000,600,10,0.5)	0.1485	0.1485	0.1485	0.9592	0.9450	0.9594
(1000,600,10,0.25)	0.1473	0.1471	0.1473	0.9583	0.9600	0.9583
(1000,600,10,0.1)	0.1518	0.1521	0.1518	0.9602	0.9850	0.9597
(1000,600,30,0.5)	0.1528	0.1525	0.1528	0.9657	0.9617	0.9659
(1000,600,30,0.25)	0.1527	0.1528	0.1527	0.9680	0.9750	0.9676
(1000,600,30,0.1)	0.1590	0.1588	0.1590	0.9711	0.9600	0.9717

Table 1: Replicated results for synthetic data. Results correspond to 95% confidence intervals.

Configuration \ Measure	ℓ	ℓ_S	ℓ_{S^c}	$\widehat{\mathrm{Cov}}$	$\widehat{\mathrm{Cov}}_S$	$\widehat{\mathrm{Cov}}_{S^c}$
(1000, 600, 10, 0.5)	0.1870	0.1834	0.1870	0.9766	0.9600	0.9767
(1000, 600, 10, 0.25)	0.1757	0.1780	0.1757	0.9810	0.9000	0.9818
(1000, 600, 10, 0.1)	0.1809	0.1823	0.1809	0.9760	1	0.9757
(1000, 600, 30, 0.5)	0.2107	0.2108	0.2107	0.9780	0.9866	0.9777
(1000, 600, 30, 0.25)	0.1956	0.1961	0.1956	0.9660	0.9660	0.9659
(1000, 600, 30, 0.1)	0.2023	0.2043	0.2023	0.9720	0.9333	0.9732

Table 2: Simulation results for the synthetic data described in (Javanmard and Montanari, 2014). The results corresponds to 95% confidence intervals.

Empirical Analysis: Simulation Study

	Javanmard & Montanari		Multisample-splitting		Ridge projection estimator		LASSO projection estimator	
Configuration	FP	TP	FP	TP	FP	TP	FP	TP
(1000,600,10,0.5)	0.0514	1	0	1	0.0420	1	0.0488	1
(1000,600,10,0.25)	0.0547	1	0	1	0.0385	1	0.0490	1
(1000,600,10,0.1)	0.0495	0.7700	0	0	0.0454	0.4750	0.0508	0.8550
(1000,600,30,0.5)	0.0531	1	0	1	0.0352	1	0.0418	1
(1000,600,30,0.25)	0.0518	1	0	0.9967	0.0413	0.9967	0.0433	1
(1000,600,30,0.1)	0.0363	0.7017	0	0.0200	0.0384	0.4383	0.0479	0.8217

Table 3: Replicated simulation results for false postive rates (FP) and and true positive rates computed at significance level $\alpha = 0.05$.

	Javanmard & Montanari		Multisample-splitting		Ridge projection estimator	
Configuration	FP	TP	FP	TP	FP	TP
(1000, 600, 10, 0.5)	0.0452	1	0	1	0.0284	0.8531
(1000, 600, 10, 0.25)	0.0393	1	0	0.4	0.02691	0.7506
(1000, 600, 10, 0.1)	0.0383	0.8	0	0	0.2638	0.6523
(1000, 600, 30, 0.5)	0.0433	1	0	1	0.0263	0.8700
(1000, 600, 30, 0.25)	0.0525	1	0	0.4	0.2844	0.8403
(1000, 600, 30, 0.1)	0.0402	0.7330	0	0	0.2238	0.6180

Table 4: Simulation results for the synthetic data described in (Javanmard and Montanari, 2014). The false positive rates (FP) and the true positive rates (TP) are computed at significance level $\alpha = 0.05$.

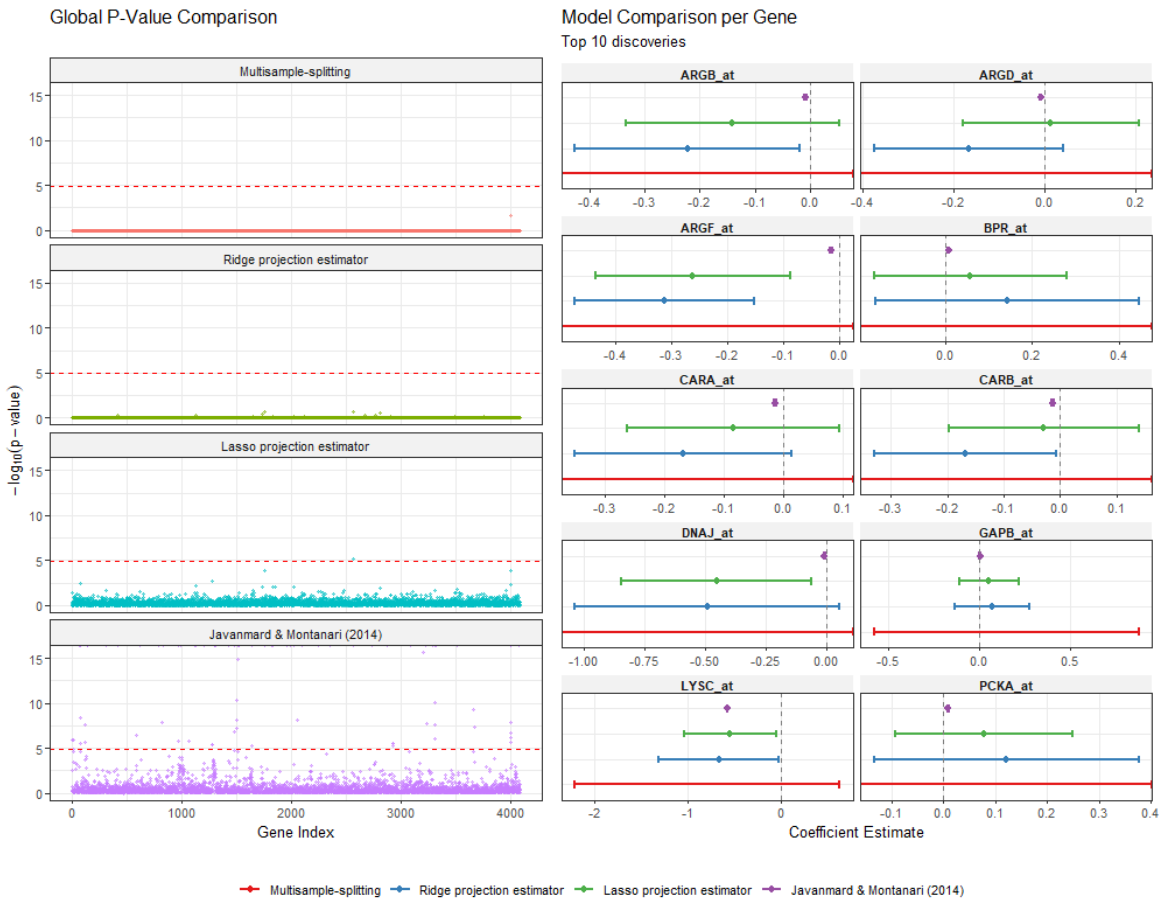
Empirical Analysis: Data Analysis

Dataset Overview (Bühlmann et al., 2014)

- **Dimensions:** $n = 71$ samples, $p = 4,088$ covariates (genes).
- **Variables:** Linear model predicting log-transformed production rate using log-transformed gene expression levels.

Empirical Analysis: Data Analysis

Takeaway: The debiased LASSO (purple) achieves the highest precision, yielding the narrowest confidence intervals and isolating signal where projections fail.



Conclusions & Future Directions

- **Theoretical Advancement:** The debiased LASSO ($\hat{\theta}^u$) successfully corrects ℓ_1 shrinkage bias, establishing a rigorous foundation for classical inference (p-values, confidence intervals) in $p \gg n$ settings.
- **The Empirical Divide:**
  - *Idealized Simulations:* The Lasso projection estimator preforms better when covariance structures are simple and well-behaved (as stated explicitly by (Zhang and Zhang, 2014))
  - *Real-World Application:* Javanmard & Montanari's optimized debiasing proves far superior at isolating signals within the complex, noisy correlation networks of actual genomic data.
- **Recommendations for Future Research:** Benchmarking must move away from overly simple circulant matrices. Future studies should intentionally simulate highly correlated data that stresses compatibility conditions to truly test these estimators.

References

Bühlmann, P., Kalisch, M., and Meier, L. (2014). High-dimensional statistics with a view toward applications in biology. *Annual Review of Statistics and Its Application*, 1(1):255–278.

Javanmard, A. and Montanari, A. (2014). Confidence intervals and hypothesis testing for high-dimensional regression. *Journal of Machine Learning Research*, 15(82):2869–2909.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B (Statistical Methodology)*, 58(1):267–288.

Zhang, C.-H. and Zhang, S. (2014). Confidence intervals for low dimensional parameters in high dimensional linear models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 76(1):217–242.

Appendix - Proof Sketch of Theorem 6 I

- **Step 1: Decompose the Estimation Error**
  Substituting $Y = X\theta_0 + W$ into the debiased estimator isolates the bias (Δ) and zero-mean Gaussian noise (Z):

$$(\hat{\theta}^* - \theta_0) = \frac{1}{\sqrt{n}}\Delta + \frac{1}{\sqrt{n}}Z$$

(where $\Delta = \sqrt{n}(M\hat{\Sigma} - I)(\theta_0 - \hat{\theta}^n)$ *and* $Z = \frac{1}{\sqrt{n}}M\mathbf{X}^\top W$*)*

- **Step 2: Bound the Bias Term (Δ)**
  Using the Triangle Inequality, we bound the infinity norm:

$$\|\Delta\|_\infty \leq \sqrt{n}\|M\hat{\Sigma} - I\|_\infty \|\hat{\theta}^n - \theta_0\|_1 = \sqrt{n}\mu_*(\mathbf{X}; M)\|\hat{\theta}^n - \theta_0\|_1$$

Appendix - Proof Sketch of Theorem 6 II

- **Step 3: Apply High-Probability Bound**
  Setting $\lambda \asymp \sigma\sqrt{(K^2 \log p)/n}$ and $e^{t^2/2} = p^{-c_0}$ bounds the ℓ_1-error. Substituting this into our bias bound yields the final claim:

$$\mathbb{P}\left(\|\Delta\|_\infty \geq \frac{4cs_0}{\phi_0^2}\sqrt{\log p}\right) \leq 2p^{-c_0}$$

Appendix - Proof Sketch of Theorem 8 I

Let $\Delta_0 \equiv \left(\frac{16ac\sigma}{C_{\min}}\right)\frac{s_0 \log p}{\sqrt{n}}$. We bound $\mathbb{P}(\|\Delta\|_\infty \geq \Delta_0)$ by conditioning on high-probability events $\mathcal{E}_n$ and $\mathcal{G}_n$.

- **Step 1: Probability Decomposition**
  By the law of total probability and the union bound:

$$\mathbb{P}(\|\Delta\|_\infty \geq \Delta_0) \leq \mathbb{P}(\{\|\Delta\|_\infty \geq \Delta_0\} \cap \mathcal{E}_n \cap \mathcal{G}_n) + \mathbb{P}(\mathcal{E}_n^c) + \mathbb{P}(\mathcal{G}_n^c)$$

- **Step 2: Bound the Complements (Bad Events)**
  By Theorem 7, the probability of the design/noise assumptions failing is exponentially small:

$$\mathbb{P}(\mathcal{E}_n^c) + \mathbb{P}(\mathcal{G}_n^c) \leq 4e^{-c_1 n} + 2p^{-c_2}$$

Appendix - Proof Sketch of Theorem 8 II

- **Step 3: Condition on Good Events**
  Conditioning on $\mathbf{X} \in \mathcal{E}_n \cap \mathcal{G}_n$, Theorem 6 controls the core probability:

$$\mathbb{P}(\{\|\Delta\|_\infty \geq \Delta_0\} \cap \mathcal{E}_n \cap \mathcal{G}_n) \leq 2p^{-\tilde{c}_0}$$

- **Step 4: Final Assembly**
  Summing the components and bounding by the dominant rate yields:

$$\mathbb{P}(\|\Delta\|_\infty \geq \Delta_0) \leq 4e^{-c_1 n} + 4p^{-\min(\tilde{c}_0, c_2)}$$

Proof Sketch - Lemma 10 I

- **Step 1: Standardize and Decompose**
  We split the standardized error into a pure Gaussian noise term $\tilde{Z} \sim \mathcal{N}(0,1)$ and a normalized bias term:

$$\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} = \frac{\sigma}{\hat{\sigma}}\tilde{Z} + \frac{\Delta_i}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}}$$

- **Step 2: Perturbation Bound**
  For an arbitrary $\varepsilon > 0$, we bound the target probability by accounting for small deviations in the bias and the variance estimator:

$$\mathbb{P}(\cdot \leq x) \leq \Phi(x + 2\varepsilon + \varepsilon|x|) + \mathbb{P}\left(\text{Scaled Bias} \geq \varepsilon\right) + \mathbb{P}\left(\left|\frac{\hat{\sigma}}{\sigma} - 1\right| \geq \varepsilon\right)$$

Proof Sketch - Lemma 10 II

- **Step 3: Apply Asymptotic Limits**
  Since the variance estimator is consistent, the absolute difference from $\Phi(x)$ as $n \to \infty$ depends only on the bias and Φ's modulus of continuity, $\delta_\varepsilon(x)$:

$$\left| \mathbb{P}\left(\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \leq x \right) - \Phi(x) \right| \leq \mathbb{P}\left(\frac{|\Delta_i|}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \geq \varepsilon \right) + \delta_\varepsilon(x)$$

- **Step 4: Uniform Convergence**
  By Theorem 8, the bias term vanishes. Since $\varepsilon > 0$ is arbitrary, letting $\varepsilon \to 0$ shrinks $\delta_\varepsilon(x) \to 0$, yielding the final claim uniformly over the sparsity class s_0:

$$\limsup_{n\to\infty} \sup_{\|\theta_0\|_0 \leq s_0} \mathbb{P}\left(\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \leq x \right) \leq \Phi(x)$$